

Luyao Xia

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Location: Auckland, New Zealand

Profile

I am a Postdoctoral Fellow at the Auckland Bioengineering Institute, The University of Auckland. My research sits at the intersection of artificial intelligence, medical robotics, and biomedical engineering. I received my Ph.D. in Control Science and Engineering from Tongji University in 2025.

Research Interests

Embodied AI, medical robotics, multimodal perception, causal inference, human-robot interaction, 3D spatial understanding, robotic perception, soft tissue segmentation, deformation tracking, force-augmented visual tracking, and mixed-reality surgical guidance.

Education

- Ph.D., Control Science and Engineering, Tongji University, China, 2025.
- M.E., Control Engineering, Tongji University, China, 2020.
- B.E., Electrical Engineering and Automation, China University of Mining and Technology, China, 2015.

Professional Experience

- Postdoctoral Fellow, Auckland Bioengineering Institute, The University of Auckland, New Zealand, July 2025 - Present.
- Research Assistant, Tongji University, China, September 2020 - January 2021.
- AI Engineer, China UnionPay, China, July 2019 - October 2019.
- R&D; Engineer, KingClean, China, July 2015 - April 2016.

Selected Publications

- **ALM-MTA: Front-Door Causal Multi-Touch Attribution Method for Creator-Ecosystem Optimization.** Y. Liu, L. Xia, H. Liu, J. Liang, H. Li, K. Gai, and H. Bai. International Conference on Learning Representations (ICLR), 2026.
- **Direct Routing Gradient (DRGrad): A Personalized Information Surgery for Multi-Task Learning Recommendations.** Y. Liu, Y. Miao, and L. Xia. AAAI Conference on Artificial Intelligence, 2025.
- **Augmented reality and indoor positioning based mobile production monitoring system to support workers with human-in-the-loop.** L. Xia, J. Lu, Y. Lu, H. Zhang, Y. Fan, and Z. Zhang. Robotics and Computer-Integrated Manufacturing, 2024.
- **Semantic knowledge-driven A-GASeq: A dynamic graph learning approach for assembly sequence optimization.** L. Xia, J. Lu, Y. Lu, W. Gao, Y. Fan, Y. Xu, and H. Zhang. Computers in Industry, 2024.
- **Construction and application of smart factory digital twin system based on DTME.** L. Xia, J. Lu, H. Zhang, M. Xu, and Z. Li. The International Journal of Advanced Manufacturing Technology, 2022.

Selected Honors and Grants

- Gold Reviewer Reward, International Conference on Machine Learning (ICML), 2026.
- Research Grant from Beijing Dajia Internet Information Technology Co., Ltd., PI, 2026-2027.
- ABI Rapid-test project fund, PI, Force-Augmented Visual Tracking for Soft Tissue Deformation in Mixed-Reality Surgical Guidance, 2025.
- Excellent Doctoral Dissertation, Tongji University, 2025.

Research Outputs

- 6 peer-reviewed journal articles and 16 peer-reviewed conference proceedings listed in the New Zealand Standard CV.

- 2 granted China patents and 3 issued software copyrights.

Public web CV. Last updated: May 2026.